Where a Quantum Reservoir Works: A Transferable Operating Band

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***Abstract*—In quantum reservoir computing, a fixed quantum system transforms an input signal, and learning is reduced to training a simple linear readout on its measured outputs. Because the quantum dynamics are never optimized, the method is appealing for today's hardware—but it leaves a basic question open: what settings actually make a fixed quantum system useful for learning, rather than just complex? We answer this for a dissipative gate-model reservoir by mapping performance over three physical control parameters: the input-rotation strength β, the nearest-neighbor ZZ coupling strength λ, and the amplitude-damping rate γ. Good performance does not appear randomly across these settings; it concentrates in a single, well-defined operating region. Crucially, this region transfers—a region identified on one set of tasks predicts the best settings for new tasks and new reservoir configurations. We confirm the region is real by showing it collapses when we remove the mechanisms that create it, and we show it can be located cheaply, in advance, using a simple memory diagnostic. Reservoir tuning thus becomes a search for a transferable, memory-preserving regime rather than a hunt for one benchmark-specific optimum.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Introduction

Reservoir computing rests on an economical idea: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional trace of its recent history, and learning reduces to fitting a single linear readout on top of that trace [1]–[4]. The dynamics are never optimized, only observed. This economy is precisely what makes the approach appealing for near-term quantum hardware: native quantum evolution can serve as the reservoir and only the classical readout must be trained [5]–[8]. Yet the same economy relocates the burden of the method onto the fixed dynamics, and so onto a question that is easy to state and surprisingly hard to answer: what makes those dynamics useful?

Recent QRC work now spans noisy gate-model devices, continuous-variable and oscillator reservoirs, weak-measurement protocols, photonic reservoirs, noise-engineered reservoirs, and experimental optical memory control [9]–[14]. Classical reservoir theory has long since refused the obvious answer. Usefulness is not maximal expressivity. A reservoir works by holding a balance: retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [4], [15]. A quantum reservoir inherits this lesson but complicates it, because these ingredients are no longer free dials. They are consequences of distinct physical resources: how strongly the input perturbs the carried state, how coherently the system mixes that state across its parts, and how it is allowed to dissipate [7], [16], [17].

We take this as an organizing principle rather than a caveat. For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask when the reservoir works, and we map it. The picture that emerges is not a single tuned operating point but a connected region—gentle input, nonmaximal coupling, controlled damping—in which good performance concentrates and from which it does not stray arbitrarily across tasks. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and a memory-based diagnostic that locates it before any task-specific search.

II. Protocol

A. Reservoir

The reservoir processes a scalar time series without resetting its density matrix, so the carried state stores recent history. We write $\rho_{t,\ell}$ for the quantum state after input time t and layer ℓ; L is the final layer, so $\rho_{t-1,L}$ is the state carried over from the previous time step. At step t, the scalar input value u_t perturbs this carried state through the same Y rotation on every qubit,

$$\rho_{t,0} = U_{\mathrm{in}}(u_t)\rho_{t-1,L}U_{\mathrm{in}}^{\dagger}(u_t), \qquad U_{\mathrm{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

Here $R_y(\cdot)$ is a single-qubit rotation, $\bigotimes_i$ means that the rotation is applied to all qubits, and † denotes the inverse conjugate transpose. The input strength β controls how strongly each step writes into the carried state.

The perturbed state then passes through identical layers: a frozen local mixer U_{mix}, a ring ZZ coupling unitary U_{λ}, and a dissipative channel $\mathcal{D}_{\gamma}$ applied independently to all n qubits,

$$\begin{aligned} \rho_{t,\ell} &= \mathcal{D}_{\gamma}^{\otimes n}\left[U_{\lambda}U_{\mathrm{mix}}\rho_{t,\ell-1}U_{\mathrm{mix}}^{\dagger}U_{\lambda}^{\dagger}\right], \\ U_{\lambda} &= \prod_{(i,j)\in E_{\mathrm{ring}}} e^{-i\lambda Z_i Z_j}. \end{aligned} \tag{2}$$

In this expression, E_{ring} is the set of nearest-neighbor edges on the qubit ring, Z_i is the Pauli-Z operator on qubit i, λ sets the coupling strength, and γ sets the damping rate. We vary only $\theta = (\beta, \lambda, \gamma)$, where β and λ are angles in radians and $\gamma \in [0, 1]$ is the amplitude-damping probability; all other gates are frozen and only a linear readout is trained. Each control

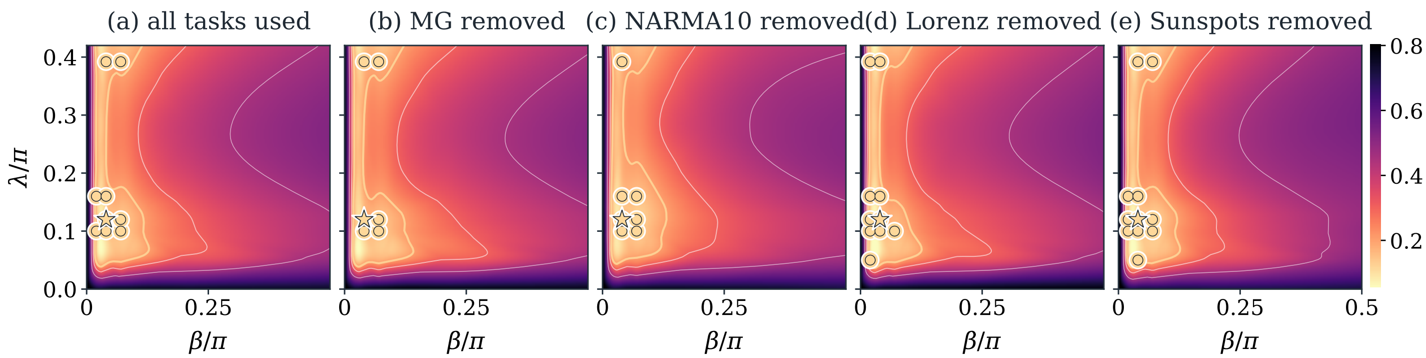

Fig. 1. Validation phase maps at $\gamma = 0.12$, the first damping slice where the strict validation-defined band is populated. The full γ scan is shown in Fig. 2; this slice is used only for a readable 2D transfer view. The first panel uses all four tasks. Each following panel recomputes the map after removing one task from the validation aggregate. Gold circles mark coordinates that repeatedly fall in the top validation quintile; the star marks the primary-band medoid projected onto this slice.

corresponds to one ingredient required by reservoir theory: β governs how new information competes with old memory, λ generates nonlinear mixing of past inputs, and γ provides the controlled forgetting on which fading memory depends.

This is a minimal testbed, not an optimized model. Four qubits and two layers keep the phase grid searchable, the ring coupling still gives recurrent many-body mixing, and the modest $Z + ZZ$ readout observes local and nearest-neighbor correlations rather than the full density matrix.

B. Configuration

Unless stated otherwise, we use four qubits, two layers, ring coupling, uniform input injection, amplitude damping, and a readout from single-qubit Z and ring-pair ZZ expectations. Training is ridge regression on these features alone. The finite search space is $\Theta_{\text{grid}} = \mathcal{B} \times \mathcal{L} \times \mathcal{G} \subset [0, \pi/2] \times [0, 0.42\pi] \times [0, 1]$, with $\mathcal{B}/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\mathcal{L}/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, and $\mathcal{G} = \{0, 0.01, 0.05, 0.12, 0.22, 0.30\}$ over seeds 42–61. Hyperparameters are selected on validation data only; holdout data never define the operating region. The resulting balanced panel contains $4 \times 20 = 80$ task-seed replicates for every grid coordinate. This count is fixed by design, not by the observed results: it is the complete factorial combination of the four benchmark tasks and twenty independent frozen-reservoir seeds, chosen to support leave-one-task and leave-one-seed transfer checks while keeping the full grid and robustness run tractable.

C. Tasks, Metric, and Splits

We use four scalar one-step prediction tasks: Mackey–Glass chaos [18], the next x coordinate of an RK4 Lorenz trajectory [19], NARMA10 memory prediction [20], and annual Sunspots from `statsmodels`. Mackey–Glass, Lorenz, and NARMA10 use washout/train/validation/holdout lengths $300/1600/400/400$; Sunspots uses $20/150/50/60$. Inputs are scaled to $[-1, 1]$ and targets are standardized. The error is normalized mean-squared error, $\text{NMSE} = \langle (y - \widehat{y})^2 \rangle / (\text{Var}(y) + 10^{-12})$, computed separately on validation and holdout splits. Ridge is selected only by validation NMSE from $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$.

D. Operating Band

To locate the useful region we rank performance within each task j and seed s. A setting $\theta \in \Theta_{\text{grid}}$ receives validation percentile rank $r_{j,s}(\theta)$, with lower rank better. The operating band contains settings that land in the top fraction p for at least a fraction q of task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

This frequency-level definition asks whether a region behaves well repeatedly, not whether one coordinate wins every task. It is weaker, and more useful, than demanding a universal optimum: it treats the band as a search prior over settings that often work. In leave-one-task or leave-one-seed tests, the band is built from the remaining replicates and scored on the withheld replicate. In the holdout audit, the validation-defined band is frozen first and evaluated once on holdout ranks and NMSE. Confidence intervals are nonparametric bootstrap percentile intervals for the mean with 50,000 resamples and a fixed seed. Because ranks are formed within each task-seed replicate before aggregation, no large-error task or difficult seed can dominate the band.

III. Results

Across the control space, low validation rank does not spread arbitrarily; it gathers into a compact, connected region rather than a scatter of isolated optima (Fig. 1). The damping coordinate is selected by validation, not fixed by hand: the near-zero $\gamma = 0.01$ slice is a weak contrast, while the strict $B_{20,0.7}$ core appears for $\gamma = 0.12$–0.30 with medoid $\gamma = 0.22$ (Fig. 2). We use $\gamma = 0.12$ in Fig. 1 because it is the first populated band slice, not because it was chosen as a magic operating point. A region that merely fit the four tasks jointly would prove little, so we test whether it survives the removal of the information it was built from. It does: when one task is withheld, the band built from the rest scores mean held-out

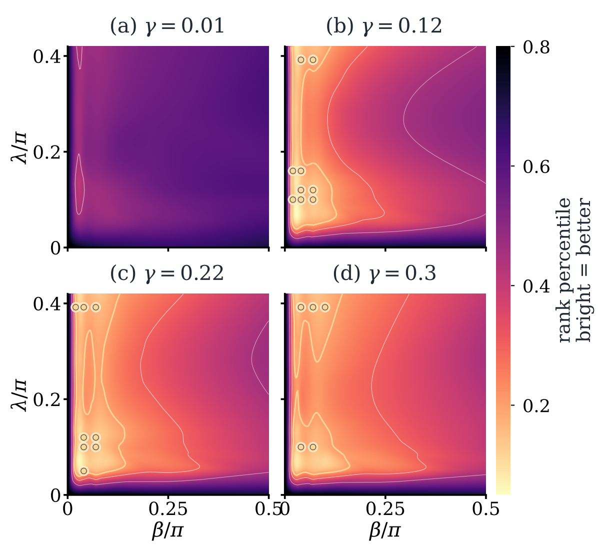

Fig. 2. Damping-slice check for the same validation-ranked grid. Bright color indicates better validation rank. Gold circles mark strict operating-band coordinates in each slice. The near-zero damping slice at $\gamma = 0.01$ provides a contrast, while the band appears across stronger nonzero damping slices.

TABLE I
HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. MEANS ARE OVER TWENTY SEEDS PER TASK; "ALL" AVERAGES THE 80 TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	7.05e-05	5.26e-05	0.104
Lorenz	1.51e-05	1.52e-05	0.136
NARMA10	0.560	0.607	0.204
Sunspots	0.209	0.209	0.168
All	0.192	0.204	0.153

rank 0.193 with CI $[0.176, 0.210]$; when one seed is withheld, the corresponding mean rank is 0.151 with CI $[0.139, 0.163]$.

The validation-defined band also holds on data it never touched: after freezing the band, a single chronological holdout evaluation remains top-quartile (Table I). A nearby four-qubit, three-layer variant preserves the phenomenon, with overlap reported against the base control region (Table II). Mechanism ablations then test causality. Removing damping, removing recurrent mixing, or replacing amplitude damping with dephasing erases the band, with retention values 0.00, 0.00, and 0.00 (Fig. 3). The collapse occurs where the base band sits, as expected if the regime is held in place by write-without-overwrite input, nonlinear mixing, and usable memory. Finally, the region can be found before task-specific search: memory capacity correlates strongly with validation rank, $\rho_s = -0.88$, as do information-processing capacities [15], whereas raw feature diversity does not (Fig. 4).

IV. Discussion and Scope

For this reservoir family, useful learning is organized by a transferable operating regime rather than by task-specific tuning. Four tasks spanning smooth chaos, nonlinear memory, and irregular real-world data obey the same region, and the region holds when any one task or seed is withheld. The useful region is not picked out by Hilbert-space size or raw feature diversity but by memory, suggesting that drive, coupling, and dissipation keep memory and nonlinearity in the balance classical reservoir theory identified [4], [15]. This is also why the ablations bite: strip out damping or mixing and the regime does not merely weaken, it disappears. The scope is deliberately narrow: all results are simulated, on one architecture family, with measurement treated as free, so finite-shot noise, calibration drift, and readout cost remain outside the claim [11], [21]. We claim neither quantum advantage nor hardware efficiency; we claim that the operating regime is the right object to study, that it transfers, and that memory helps find it.

TABLE II
ARCHITECTURE ROBUSTNESS AFTER VALIDATION-ONLY BAND SELECTION. OVERLAP IS JACCARD OVERLAP WITH THE BASE BAND IN THE SHARED (β, λ, γ) GRID.

Setting	Band?	Held-out rank	Overlap
4q, 2 layers, Z+ZZ	yes	0.153	1.00
4q, 3 layers, Z+ZZ	yes	0.154	0.04

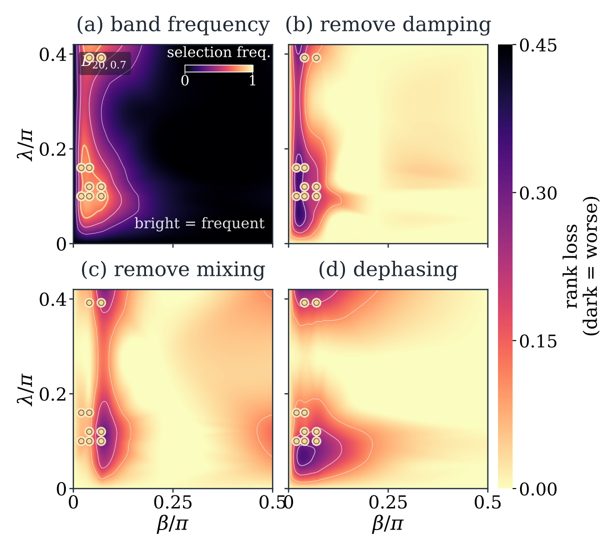

Fig. 3. Why the operating band is meaningful. (a) Bright coordinates are frequently selected by validation. (b–d) Removing damping, removing mixing, or replacing amplitude damping by dephasing concentrates rank loss around the same gold band, showing that the useful region depends on these mechanisms.

V. Conclusion

The usefulness of a dissipative quantum reservoir is not a property of size alone but of where it is operated. Across tasks and reservoir initializations, strong performance gathers in a transferable region where input drive, recurrent coupling, and dissipation hold memory and nonlinearity in balance. The regime transfers, collapses under targeted mechanism removals, and can be found in advance by memory-based diagnostics rather than raw feature diversity. The result is intentionally limited: all experiments are simulated, measurement is treated as free, and finite shots, calibration drift, device noise, and readout cost remain outside the present claim.

Reproducibility appendix.

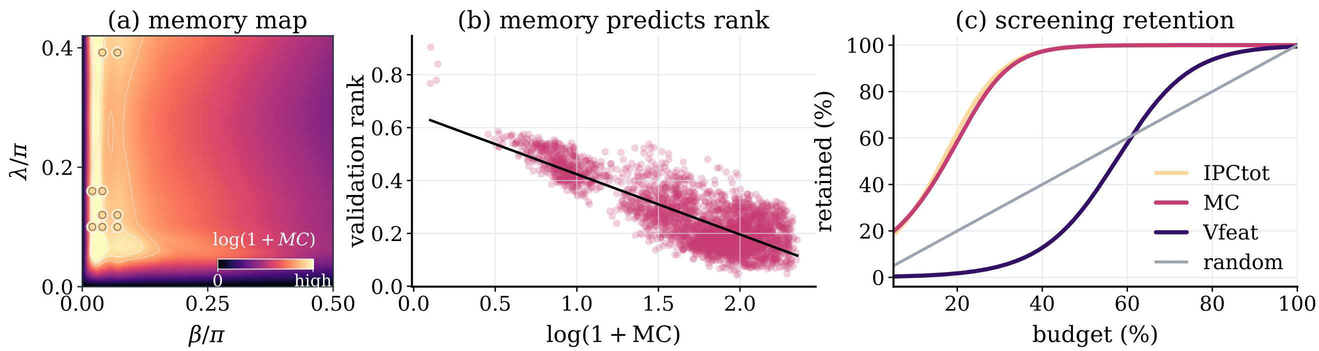

Fig. 4. Memory-based diagnostics identify the operating region before task-specific search. (a) The task-free memory map shows $\log(1 + \mathrm{MC})$ on the same $\gamma = 0.12$ control slice as the phase maps. The gold operating-band points lie inside the structured memory-rich region, showing that the band is visible from reservoir dynamics alone. (b) Across all sampled settings and seeds, higher memory predicts lower validation rank, giving a global quantitative version of the same effect. (c) Screening efficiency: the fraction of genuinely good reservoirs retained as a function of how many settings are kept. Screening by memory or information-processing capacity recovers the useful settings far earlier than random selection, while raw feature diversity remains much weaker.

Package. The project repository contains the code, checked-in CSV/JSON results, and final figures used in this paper.

Frozen reservoir. The simulator evolves the full density matrix exactly, with no finite-shot sampling. It starts from $\rho_0 = |0\cdots0\rangle\langle0\cdots0|$ and carries state between time steps without reset. For each seed, `default_rng(s)` draws one fixed mixer $U_{\mathrm{mix}} = \bigotimes_i R_x(\theta_i)$, $\theta_i \sim U[0, 2\pi)$, reused at every step and layer. Each layer applies $U_\lambda U_{\mathrm{mix}}$ on ring edges, then local amplitude damping. The readout is ridge regression on exact Z_i and ring-pair Z_iZ_j expectations with an unpenalized intercept.

Selection and diagnostics. The (β, λ, γ) grid and task splits are in Sec. II. Ridge penalties $\{10^{-8}, \ldots, 10^3\}$ are chosen by validation NMSE only; holdout data are used only after the band is fixed. MC/IPC diagnostics use an independent $U[-1, 1]$ drive of length 900, washout 150, ridge $\alpha = 10^{-8}$, and a 70/30 chronological split. MC sums held-out positive R^2 over delays 1–10. IPC adds linear delays 1–20, quadratic Legendre delays 1–10, and ten cross-delay products, clipping negative R^2 to zero. CIs are nonparametric bootstrap percentile intervals with 50,000 resamples and a fixed seed.

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